



CluRF: Combining Unsupervised Clustering with Explainable Ensemble Learning for Risk Prediction

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Abstract. Cardiovascular diseases (CVDs) remain the leading cause of mortality worldwide, highlighting the critical need for accurate and interpretable early risk prediction. While conventional tools such as the Framingham Risk Score offer high transparency, they rely on rigid assumptions and fixed variable sets, limiting their adaptability to patient-specific variations. Recent machine learning approaches improve predictive performance but often lack interpretability, hindering clinical acceptance. To address these limitations, we propose CluRF, a hybrid framework that combines unsupervised clustering with Random Forest classifiers and integrates SHAP-based explanations. CluRF partitions the population into latent subgroups based on feature similarity, enabling personalized modeling within each cluster and providing global and individualized interpretability. We evaluate our method on four real-world heart disease datasets and demonstrate that CluRF achieves competitive or superior performance compared to state-of-the-art models in terms of F1-score, accuracy, and AUC. Importantly, it maintains a lightweight and explainable structure suitable for clinical deployment.

Keywords: Heart Attack Risk · Machine Learning · SHAP · Ensemble Learning · Unsupervised Clustering

1 Introduction

Cardiovascular diseases (CVDs) have remained the leading cause of death worldwide, accounting for approximately 17.9 million deaths annually¹. Among them, acute myocardial infarction (AMI), or heart attack, is one of the most severe and sudden cardiovascular events. A heart attack can cause permanent cardiac tissue damage or even sudden death, often without any prior symptoms. According to clinical reports, many patients do not receive timely intervention

¹ Cardiovascular diseases (CVDs). Published by the World Health Organization. Available at: [https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-\(cvds\)](https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-(cvds)).

because the onset is either misinterpreted or goes undetected until it becomes life-threatening.

While treatment strategies have advanced, ranging from stent implantation to thrombolysis and pharmacological therapy [1, 2], these methods are fundamentally reactive in nature. They are administered after a heart attack has already occurred. In contrast, proactive risk prediction offers a paradigm shift: enabling healthcare providers to identify high-risk individuals early and provide timely lifestyle or medication interventions. However, despite the increasing availability of electronic health records (EHR), biometric signals from wearable devices, and population-scale medical datasets, accurate and interpretable heart attack prediction remains a major research challenge, particularly for personalized care and real-time application.

This has led to significant efforts in computational health and machine learning communities to develop models capable of integrating diverse risk factors and predicting AMI before onset. A robust and clinically deployable model must not only be accurate but also interpretable, computationally efficient, and capable of handling heterogeneous patient data.

The evolution of heart attack prediction models can be traced from simple rule-based clinical calculators to modern data-driven machine learning frameworks. Early tools such as the Framingham Risk Score [3] and ASCVD calculators [4] estimated risk using linear combinations of a few major variables like age, cholesterol, and smoking history. While valuable in preventive cardiology, these methods lacked flexibility and could not model complex interactions.

Traditional statistical models, including logistic regression and linear discriminant analysis (LDA), have long supported clinical risk prediction due to their structured and transparent decision-making processes. Their reproducibility made them a staple in medical research, yet they rely on assumptions—such as linear relationships and feature independence—that are rarely valid in real-world clinical data, which often involves complex interdependencies and nonlinear patterns. In response to these limitations, the past two decades have seen increased adoption of machine learning methods for heart disease prediction. Techniques such as support vector machines (SVMs) [5], random forests (RFs) [6], k-nearest neighbors (KNN) [7], and neural networks (NNs) [8] offer the ability to model nonlinear, high-dimensional relationships and latent interactions within diverse clinical datasets.

Despite notable performance improvements, these machine learning approaches face persistent challenges. First, many models overlook population heterogeneity, assuming uniformity across patient groups despite known demographic differences in risk presentation. Second, the lack of interpretability—especially in deep learning models—limits their clinical utility, as physicians need understandable justifications for predictions. Third, deployment barriers such as computational complexity and extensive tuning hinder practical use in real-time healthcare settings. Recent studies [9, 10] suggest that incorporating personalization and explainability enhances trust and usability. Motivated by these insights, we propose a framework that integrates subgroup discovery via

unsupervised clustering, applies lightweight ensemble models for prediction, and uses SHAP-based explanations to make model outputs clinically interpretable.

This study aims to design a predictive framework that enhances both accuracy and interpretability in heart attack risk assessment. It focuses on four key challenges: handling population heterogeneity by identifying meaningful patient subgroups; improving model generalization through subgroup-specific classifiers; increasing interpretability with explanations at both global and individual levels; and achieving computational efficiency without relying on complex, resource-intensive models.

To tackle these issues, we introduce CluRF, a hybrid approach that combines unsupervised clustering with Random Forest (RF) classification. Patients are first grouped based on feature similarity, and separate RF models are trained within each cluster to capture subgroup-specific patterns. To support clinical reasoning, we use SHAP (SHapley Additive exPlanations) to provide interpretable insights into feature contributions at both the population and individual levels.

2 Related Work

In clinical practice, early heart attack prediction relies mainly on structured risk-scoring systems, such as the Framingham Risk Score (FRS) [3], which predicts 10-year coronary risk based on age, blood pressure, cholesterol, smoking, and diabetes; SCORE, derived from over 200000 Europeans, provides region-specific charts for fatal cardiovascular events [11]; and QRISK2, incorporating ethnicity, socioeconomic status, and comorbidities, enables more personalized risk estimates in the UK [12]. Additionally, large epidemiological cohorts like the Framingham Heart Study [13] employed multi-variable Cox proportional-hazards models to estimate 5 or 10-year coronary risk, later converted into point-based scores by Sullivan et al. [14] for ease of clinical use. However, these approaches depend on predefined variables and assume linear relationships between risk factors and outcomes, limiting flexibility and motivating recent data-driven methods.

With the rapid development of data-driven techniques in healthcare, increasingly sophisticated models have been applied to heart disease prediction tasks. These models can be broadly divided into two categories: Machine Learning (ML) and Deep Learning (DL).

Recent ML-based approaches have achieved notable progress in heart disease prediction. Canedo et al. [15] used Frequency Modulated Möbius (FMM) modeling to extract morphological features from 10-second EEG epochs, offering high interpretability by decomposing waveforms into clinically meaningful components but introducing heavy computational overhead and reliance on manual feature engineering. Garg et al. [16] demonstrated that a minimalist model using only two features (ECG and ST_slope) from a Kaggle dataset can achieve robust performance: they compared Naive Bayes (NB), Decision Table (DT), J48 (a decision tree variant), and Random Forest, finding that RF delivered the most stable and effective classification with minimal overhead, although it does not

generalize well to higher-dimensional feature sets. Collectively, these methods inspire exploration of low-dimensional, interpretable clustering strategies that balance strong predictive power with clear feature separation.

In the domain of deep learning, a notable contribution came in 2022 when Hourri et al. [17] proposed DeepSCA, which combined PatternNet neural networks with the Sine Cosine Algorithm to automatically optimize network architecture. This method demonstrated the benefit of metaheuristic tuning. In 2024, Dritsas and Trigka [18] designed a hybrid *CNN + GRU* model with SHAP explainability on the UCI Cleveland dataset. Compared to conventional deep learning models such as MLP, LSTM, and GRU alone, the *CNN + GRU* hybrid consistently delivered better results while offering interpretability through SHAP analysis. In the same year, Nawar et al. [19] applied Independent Component Analysis (ICA) as a preprocessing technique to enhance various classifiers, including KNN, RF, SVM, NB, and GB. ICA significantly improved the performance of these classifiers, especially RF and KNN.

DeepSCA exemplifies the benefits of automated architecture search, but its opaque decision process limits its deployment and clinical practicality. The CNN+GRU hybrid explicitly captures spatial - temporal structure and, offers explanation into feature contributions; however, the model exhibits reduced robustness in limited-sample regimes. An ICA-enhanced ML pipeline can improve the accuracy of conventional classifiers - most notably RF and KNN - yet the lack of standardized post-hoc tools (e.g., SHAP/LIME) limits transparency for clinician review. Taken together, these approaches delineate a familiar trade-off: higher representational capacity tend to arrive with greater computational burden and black-box characteristics, whereas traditional models still require higher accuracy to achieve clinical accountability.

Recent progress in large language models (LLMs) has shown strong promise for healthcare, particularly in handling unstructured clinical text. Models like BioGPT [20], trained on 15 million PubMed abstracts, achieved state-of-the-art results on biomedical NLP tasks such as relation extraction and question answering, while Med-PaLM 2 [21] reached expert-level accuracy (86.5%) on MedQA through advanced prompting and domain-specific tuning. However, despite their strengths in textual understanding, LLMs remain less effective for structured, tabular risk prediction tasks due to high computational demands, limited interpretability, and poor alignment with relational data patterns. These trade-offs underscore the need for future models that integrate automated or hybrid learning with transparent feature selection and explanation mechanisms to achieve both high performance and clinical usability.

In clinical settings, interpretability is essential for high-stakes tasks like heart disease diagnosis, since black-box models (e.g., neural networks or ensembles) often lack transparency. To address this, Wang et al. [9] used AutoGluon with SHAP to rank features (e.g., cholesterol, chest pain, age) per prediction - providing both global and local explanations, though SHAP can be computationally expensive and indicates correlation rather than causation. Skouteli et al. [10] paired XGBoost with rule-extraction and argumentation logic to pro-

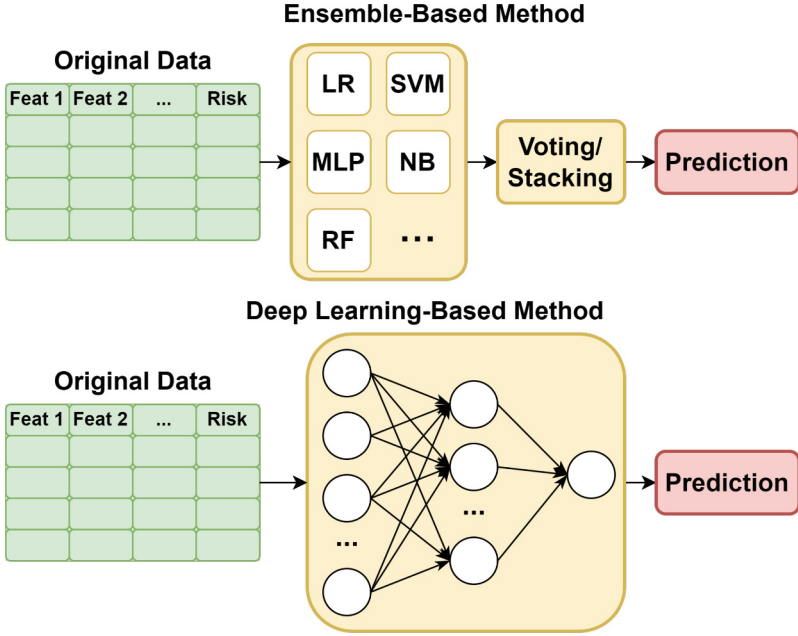


Fig. 1. Paradigms of Old Structures.

duce “IF *age* > 50 AND *cholesterol* > 240 THEN high risk” style rules, enhancing transparency but risking oversimplification and redundant rule sets. Xie et al. [22] introduced a prototype-based ECG classifier that mimics clinical analogies (“patients similar to you”), improving interpretability at the cost of nuance in rare cases and requiring careful prototype selection. Finally, Rezk et al. [23] combined SHAP and LIME on an ensemble of classifiers to highlight features like resting blood pressure and heart rate, though LIME’s perturbation-based sampling can be unstable and the dual approach increases computational overhead.

These reviewed methods highlight the need for interpretable yet efficient models like **CluRF**, which is introduced in the next section.

3 Methodology

In recent years, the prediction of heart attack risk has increasingly relied on sophisticated machine learning paradigms. As illustrated in Fig. 1, most existing approaches fall into two dominant categories: ensemble-based learning frameworks and deep learning architectures.

Ensemble techniques aims to combine the predictive capabilities of multiple base learners (e.g., SVM, Random Forest, Naïve Bayes, etc.). These methods enhance performance by aggregating diverse decision boundaries and reducing model variance. However, they typically suffer from increased complexity,

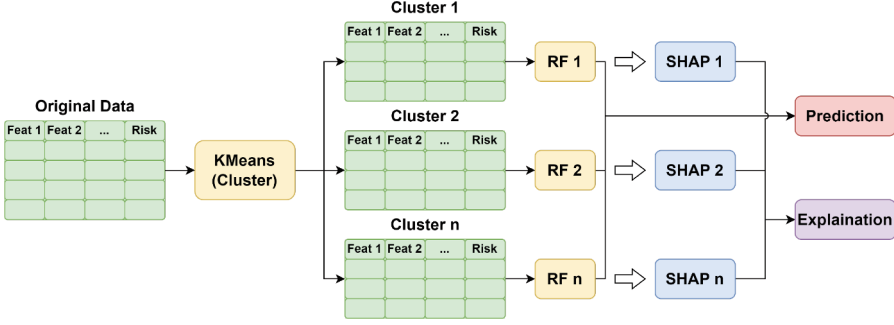


Fig. 2. The architecture overview of our method, **CluRF**.

reduced interpretability, and longer training/inference times, which may limit their suitability in time-sensitive or resource-constrained healthcare environments. Deep learning models, primarily multilayer perceptrons (MLPs), which are capable of learning non-linear relationships and complex feature interactions. These models are often used in an end-to-end fashion and can integrate a wide variety of input features. Although deep neural networks have demonstrated competitive performance, they require large amounts of training data and computational resources, and often act as “black boxes”—a drawback in high-stakes domains such as medical decision-making.

3.1 Model Structure Overview

To address the limitations of these mainstream approaches, we propose **CluRF**, a hybrid framework that combines Cluster and Random Forest pipeline. As shown in Fig. 2, by integrating unsupervised clustering prior to model training, **CluRF** can effectively partition the population into subgroups with shared characteristics, enabling more specialized and accurate risk prediction models per group.

3.2 Prediction Model

The prediction model of our proposed **CluRF** framework lies in its two-stage modeling pipeline: as shown in Fig. 2: an unsupervised clustering stage, followed by a cluster-specific prediction stage using Random Forest.

In the first stage, the dataset is split into k clusters using the KMeans algorithm, aiming to group individuals with similar features. This step is essential in capturing latent sub-populations often observed in medical data due to individual differences. By isolating these groups, we reduce the variance within each cluster and allow the subsequent model to focus on more homogeneous data distributions.

In the second stage, an independent Random Forest (RF) model is trained for each cluster. As illustrated in Fig. 3, RF model is an ensemble of decision

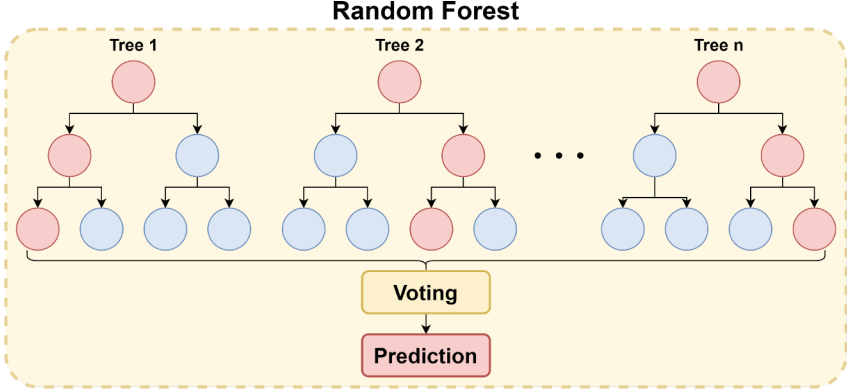


Fig. 3. Random Forest Classifier.

trees, where each tree is trained on a bootstrap sample of the data and votes for a class label, and final prediction is determined by majority voting across the trees.

This model strikes a practical balance between model capacity and interpretability: compared with simple linear models (e.g., LDA, LR), RF can capture non-linear feature interactions without heavy feature engineering; compared with complex deep models, it is lighter to tune, more data-efficient, and computationally cheaper—properties that fit clinical tabular data where sample sizes are modest and patient characteristics are heterogeneous. In summary, we choose RF due to following reasons:

(i) *Robustness to overfitting*: bagging on bootstrap samples and majority voting across trees; improve generalization, which is crucial for small or noisy medical datasets; (ii) *Interpretability*: RF provides feature-importance estimates out-of-the-box, helping identify clinically meaningful risk factors; (iii) *Efficiency and stability*: training and inference are fast on moderate-dimensional tabular data, with few sensitive hyperparameters to tune (trees/depth).

In this structure, once the input data is assigned to a cluster, the corresponding RF would predict the risk of heart attack in that cluster. This design enables each RF model to specialize in its subgroup, enhancing both predictive accuracy while keeping the efficiency.

3.3 Explanation Model

Another important part of our **CluRF** framework is model explanation. Inspired by previous studies ([9, 23]), we introduce SHAP to explain the decision-making process of each cluster-specific RF model. As shown in Fig. 4, SHAP (SHapley Additive exPlanations) provides a unified framework grounded in cooperative game theory, offering both global and local interpretability for complex machine learning models.

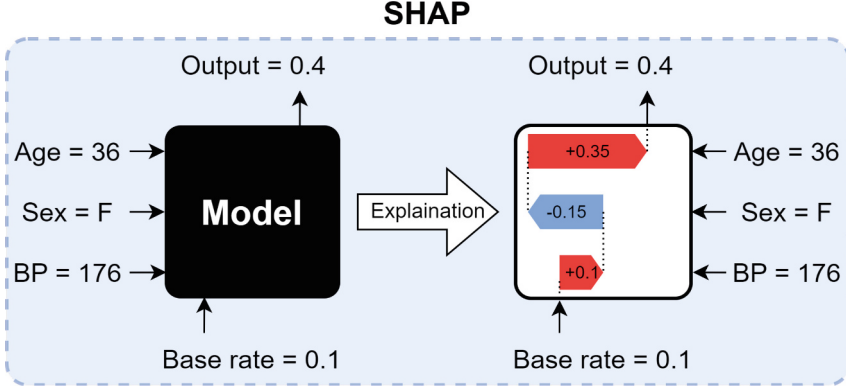


Fig. 4. SHAP structure and working principle.

SHAP assigns each feature an importance value, termed the *Shapley value* that represents its marginal contribution to the model’s output, averaged over all possible combinations of features. Formally, let $f(x)$ be the output of the model for input x , and let x_i be the i -th feature. The Shapley value ϕ_i for feature i is defined as:

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} [f(S \cup \{i\}) - f(S)], \quad (1)$$

where F is the set of all features and S is a subset of features not containing i . This formulation satisfies desirable properties such as *local accuracy* (the prediction is the sum of contributions), *consistency* (feature importance changes consistently with model behavior), and *missingness* (features not used in the model have zero contribution).

SHAP interpretation is generally categorized into two levels:

- **Global Interpretation:** SHAP *summary plots* are used to show the average magnitude and direction of feature contributions across the dataset. This helps to identify the most influential features in the model’s overall decision-making process, allowing domain experts to understand general patterns.
- **Local Interpretation:** SHAP *waterfall plots* are employed to explain individual predictions. These plots visualize how each feature contributes to increasing or decreasing the prediction for a specific sample. This fine-grained explanation helps answer questions like: ‘Why did the model predict a high (or low) risk for this patient?’

Due to its solid theoretical foundation and practical interpretability, SHAP has become one of the most widely adopted tools for explaining black-box models, especially in sensitive applications where understanding the rationale behind a prediction is as important as the prediction itself.

It is important to clarify that SHAP reveals statistical associations rather than causal relationships. SHAP values indicate how strongly features are correlated with model outputs, but they do not establish whether changing a feature would directly cause a change in prediction. In contrast, causal inference techniques explicitly model interventions and counterfactuals to uncover cause-effect mechanisms such as do-calculus [24] or modern causal discovery algorithms [25]. Thus, while SHAP is a powerful tool for interpretability, it should be complemented with causal inference approaches when the research question requires establishing causality.

Table 1. Performance comparison of different methods across four datasets. For each method, we choose the hyperparameter that could achieve best performance. Number with **bold** and underline are top and second top results.

Method	heart			heart-1			UCI-1190-11			cleveland		
	F1	ACC	AUC	F1	ACC	AUC	F1	ACC	AUC	F1	ACC	AUC
LDA	0.7831	0.7772	0.9006	0.7923	0.7902	0.8810	0.8279	0.8235	0.9085	0.6452	0.7963	0.9466
LR	0.7742	0.7717	0.9008	0.7487	0.7610	0.8790	0.8182	0.8151	0.9088	0.6452	0.7963	0.9466
RF	0.8670	0.8533	0.9215	0.9852	0.9854	0.9996	0.9430	0.9370	0.9682	0.7000	0.7778	0.8874
SVM	0.8815	0.8641	0.9484	0.9132	0.9073	0.9642	0.9051	0.8908	0.9476	0.8500	0.8889	0.9257
MLP (1 Layer)	0.8990	0.8913	0.9016	0.8053	0.7854	0.9084	0.7647	0.7647	0.5106	0.5741	0.3018	0.6375
MLP (2 Layers)	0.9029	0.8913	<u>0.9256</u>	0.8661	0.8537	0.9459	0.8775	0.8697	0.8235	0.8333	0.3590	0.8750
Stacking [26]	<u>0.9180</u>	<u>0.9100</u>	0.9200	0.9840	0.9800	0.9800	—	—	—	—	—	—
Voting [23]	—	—	—	—	—	—	0.9609	0.9622	<u>0.9575</u>	—	—	—
Hybrid [18]	—	—	—	—	—	—	—	—	—	0.8900	<u>0.9100</u>	<u>0.9500</u>
GRU	—	—	—	—	—	—	—	—	—	0.8700	0.8900	0.9300
KNN [19]	—	—	—	—	—	—	—	—	—	<u>0.9100</u>	<u>0.9100</u>	0.9600
Ours ($k \in \{3, 4\}$)	0.9073	0.8973	0.8966	<u>0.9905</u>	<u>0.9903</u>	<u>0.9906</u>	0.9421	0.9375	0.9361	0.8772	0.8833	0.8839
Ours ($k \in \{5, 6, 7\}$)	0.9223	0.9140	0.9140	1.0	1.0	1.0	<u>0.9545</u>	<u>0.9506</u>	0.9489	0.9153	0.9180	0.9186

3.4 Experiment Design

To evaluate the proposed **CluRF** model, we conduct experiments on multiple real-world mobility datasets. This section details how we set up the experiments, and the comparative results across all baselines.

Datasets: We select four open-source and widely used datasets: **heart** Dataset², **heart-1** Dataset³, **UCI-1190-11** Dataset⁴, and **cleveland** Dataset⁵,

² Heart Failure Prediction Dataset: <https://www.kaggle.com/datasets/fedesoriano/heart-failure-prediction>.

³ Heart Disease Dataset: <https://www.kaggle.com/datasets/johnsmith88/heart-disease-dataset>.

⁴ UCI-1190-11 Dataset: <https://github.com/Abdulrakeeb/Heart-disease-dataset/tree/main>.

⁵ Cleveland Dataset: <https://archive.ics.uci.edu/dataset/45/heart+disease>.

Table 2. Details of each dataset after pre-processing

Dataset	heart	heart-1	UCI-1190-11	cleveland
# Samples	918	1,025	1,190	297
# Features	10	10	8	10

which contain real heart disease records from different hospitals or clinics. We also apply some pre-processing across all datasets. In summary, for all four datasets, the process includes: (1) Drop missing samples. Since all data are from real cases, we do not fill the missing values with processed data. (2) Remove low correlative features. Specifically, we exclude features with low (< 0.25) Pearson correlation coefficient to all other features, which have little contribution to our model. (3) Vectorization and normalization. We transform categorical and textual inputs into numerical vectors, followed by normalization to standardize the feature scales. After process, the details in each dataset are shown in Table 2.

Baselines: We selected a few famous and representative methods and models as baselines: (1) Linear Discriminant Analysis (LDA). (2) Logistic Regression (LR). (3) Random Forest (RF). (4) Support Vector Machine (SVM). (5) Multilayer Perceptron (MLP). (6) Stacking [26] (Specific in **heart** and **heart-1** dataset). (7) Voting [23] (Specific in **UCI-1190-11** dataset). (8) Hybrid [18] (Specific in **cleveland** dataset). (9) KNN [19] (Specific in **cleveland** dataset).

Some baselines (e.g., KNN [19], Hybrid [18], Voting [23]) were only evaluated on specific datasets where they were originally proposed and tuned (e.g., **cleveland** or **UCI-1190-11**). Directly applying them to other datasets without re-training would result in unfair comparisons. In addition, KNN becomes inefficient and unstable on larger datasets such as **heart** and **heart-1**. For clarity, omitted results are marked with “-” in Table 1.

Evaluation Metrics: We assessed the performance of **CluRF** using F1-score, accuracy, and Area Under Curve (AUC). These metrics are formulated by the correctness of prediction. The formulas for calculating these metrics are:

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2TP}{2TP + FP + FN}, \quad (2)$$

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}, \quad (3)$$

$$AUC = \int_0^1 TPR(FPR) d(FPR), \quad (4)$$

where $TPR = \frac{TP}{TP+FN}$ is the true positive rate and $FPR = \frac{FP}{FP+TN}$ is the false positive rate.

Implementation Details: We choose Random Forest (RF) as the base classifier, with the number of estimators set to 100 and the maximum depth tuned through cross-validation. For subgroup discovery, k -means clustering is applied with varying cluster numbers $k \in \{4, 5, 6, 7\}$, as illustrate in Sect. 3.5. Each experiment is repeated with multiple random seeds to ensure robustness. The experiments were conducted on a machine with an Intel Core i9 processor, 16 GB of RAM. The software setup included Python 3.12 as the primary programming language.

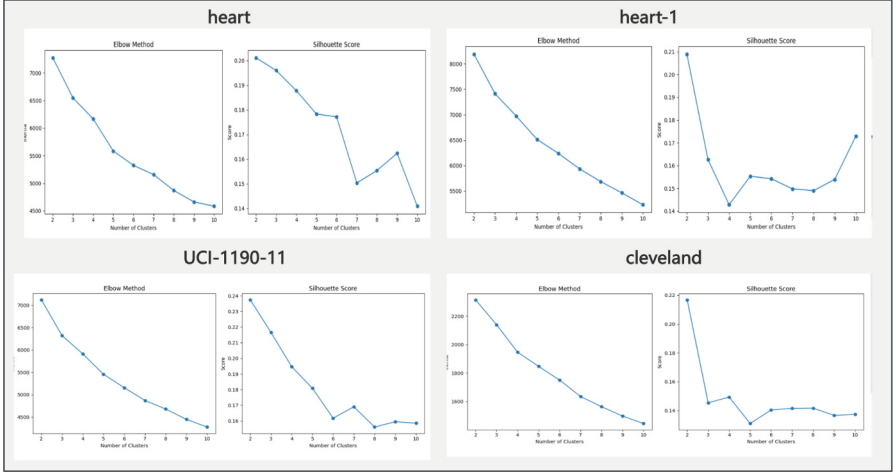


Fig. 5. Elbow Method for optimal K selection.

3.5 Prediction Result

We divide all datasets into 80% training and 20% test sets. All clustering procedures (KMeans++ fitting) were conducted exclusively on the training set. During inference, each test sample was assigned to the nearest cluster centroid without participating in the clustering process.

Table 3. Number of samples for cluster. Train* and Test* means the range of training/test samples per cluster

Dataset	heart	heart-1	UCI-1190-11	cleveland
# Cluster	6	5	7	4
Train*	105~123	120~183	112~192	58~62
Test*	27~47	31~46	29~48	15~16

To determine the optimal number of clusters K , we applied both the Elbow Method and Silhouette Score. As shown in Fig. 5, different datasets exhibit different optimal cluster numbers due to variations in data scale, feature composition, and inherent group structures. For example, the **heart** and **heart-1** dataset tends to stabilize around $K = 6$ and $K = 5$, while the larger and more heterogeneous **UCI-1190-11** dataset supports higher $K = 7$ values. In contrast, the smaller **cleveland** dataset achieves its best silhouette score at $K = 4$. This demonstrates that the choice of K must be adapted to the characteristics of each dataset rather than fixed across all scenarios. After cluster, the number of samples in each cluster is shown in Table 3.

Table 1 shows a comparison between our method and previous methods. As seen in the table, our proposed method outperforms over three dataset, and only lags behind the Voting method by merely 0.01 on the **UCI-1190-11** dataset.

First, compared with traditional machine learning models such as Random Forest and Support Vector Machine, our approach achieves a substantial performance improvement across all evaluated datasets. This advantage primarily comes from the incorporation of clustering stage, which effectively captures the vary features among individuals. By identifying and separating latent subgroups within the population, the model enables the subsequent classifiers (RF) to specialize in more homogeneous feature spaces, thereby enhancing predictive accuracy and robustness.

Second, in comparison with recent state-of-the-art approaches—most of which adopt complex ensemble learning strategies [18, 23, 26] - our method offers a favorable trade-off between model simplicity and predictive performance. While SOTA methods often involve high computational costs, **CluRF** remains structurally lightweight and more interpretable, yet achieves comparable or superior results on the **heart**, **heart-1**, and **cleveland** datasets. Even on the **UCI-1190-11** dataset, where Voting [23] shows a marginal advantage, the performance gap is no more than 0.01 in all metrics. These results underscore **CluRF**'s potential for real-world deployment.

Moreover, our ablation results further reveal that increasing the number of clusters k within a reasonable range leads to consistent performance gains, with improvements of approximately 0.010.04 observed across multiple evaluation metrics. This trend can be attributed to the fact that finer-grained clustering enables the model to group individuals with more similar features, thereby allowing each RF classifier to more precisely model intra-group patterns and better distinguish them from other populations. However, this must be balanced against the risk of over-fragmentation, which may reduce the statistical power within each cluster.

3.6 Ablation Study

To further examine the role of the clustering stage, we conduct ablation experiments by varying the contribution of number of clusters k . Figure 6 summarizes the results on **heart** and **cleveland** datasets, where F1, Accuracy, and AUC are plotted against different cluster numbers.

Across both datasets, a clear upward trend can be observed as the number of clusters increases. The metrics improve rapidly when moving from a single cluster (i.e. RF on global dataset) to a moderate number of clusters, highlighting the effectiveness of clustering subgroups. After the initial gain, the improvements become less pronounced, and the performance curves tend to flatten when the number of clusters is relatively large. Overall, the results across datasets confirm the contribution of clustering: it strengthens the model by capturing structural heterogeneity, but limited once the subgroup division becomes too dense. A moderate clustering level thus achieves a balanced trade-off between granularity and robustness.

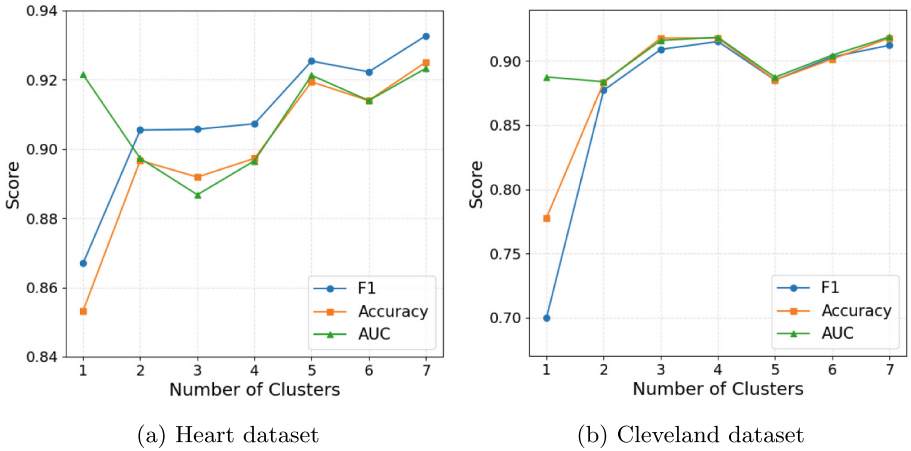


Fig. 6. Ablation study on two different dataset. Performance of clusters k in terms of F1, Accuracy, and AUC.

3.7 Model Explanation

In our **CluRF** model, we employ a two-stage approach that combines clustering with Random Forest classifiers to predict the risk of heart disease. Specifically, we first partition the dataset into subgroups using clustering, and then train a separate Random Forest model for each cluster. While this strategy improves predictive performance, it also introduces a significant limitation: the decision-making process within each model remains a “black box” that offers limited transparency to clinicians and patients - a critical concern in medical applications.

To enhance model interpretability, build trust among healthcare professionals, and help patients understand the key factors contributing to their individual heart attack risk, we integrate the **SHAP** method into our framework. In our project, we leverage two key SHAP visualization tools to interpret the predictions of our cluster-specific Random Forest models:

The *summary plot*, as shown in Fig. 7, which aggregates SHAP values across all samples within a cluster, ranking features by their overall impact on the model’s predictions. This global view helps clinicians identify which features are most influential for the population in that cluster. For example, *chest pain type*, *heart slope*, and *age* often emerge as dominant predictors, providing insight into common risk factors.

The *waterfall plot*, as illustrated in Fig. 8, explaining the prediction for an individual patient by breaking it down into the additive contributions of each feature. It visually demonstrates how certain variables push the predicted risk upward or downward compared to the model baseline, helping doctors and patients understand why a specific patient is ultimately predicted as low or high risk.

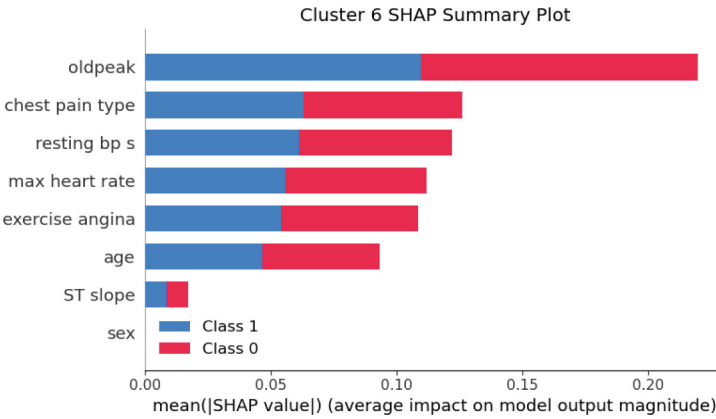


Fig. 7. An example of SHAP summary plot.

These figures demonstrate SHAP’s explanatory capability through a representative example. The patient belongs to a high-risk cluster but is predicted to have a low probability of heart attack. The waterfall plot (right) shows that *oldpeak* is the most impactful feature, reducing the predicted risk by 17%. Other influential features include *exercise-induced angina* and *resting blood pressure*, which further contribute to the final output. This decomposition provides a transparent rationale behind the model’s decision.

Beyond individual predictions, the SHAP summary plot (left) highlights the dominant features across all patients in the same cluster. If *oldpeak* frequently appears as a top contributor in high-risk groups, physicians can place greater emphasis on this feature during diagnosis. Thus, the model serves not only as a predictive tool but also as a clinical decision support system.

The integration of SHAP into the **CluRF** framework effectively addresses one of the major challenges in deploying machine learning models in healthcare: interpretability. It enables both localized (patient-level) and global (population-

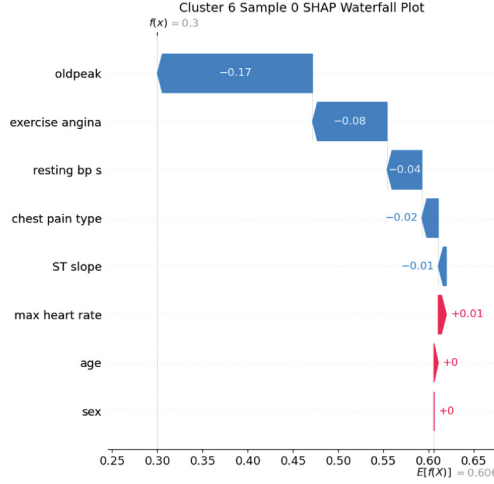


Fig. 8. An example of SHAP waterfall plot.

level) insights, fostering transparency, clinical relevance, and patient understanding.

4 Conclusion

In this study, we proposed **CluRF**, a novel and interpretable framework for heart attack risk prediction that integrates unsupervised clustering, Random Forest classifiers, and SHAP-based explanations. Our model is designed to simultaneously improve predictive accuracy and interpretability, addressing the limitations of conventional black-box machine learning models that often lack clinical transparency.

Through comprehensive experiments on four real-world heart disease datasets, **CluRF** consistently achieved top-tier performance in most evaluation scenarios, surpassing or closely matching state-of-the-art methods in terms of F1 score, accuracy and AUC. This demonstrates not only the robustness of our approach, but also its generalizability across diverse data distributions.

More importantly, **CluRF** provides strong interpretability through the use of SHAP, allowing global and individualized explanations of predictions. Unlike opaque models, our framework allows clinicians to understand why a certain prediction was made, identifying the most influential features, such as type of *chest pain*, *resting blood pressure*, or ST depression (*oldpeak*). These insights enable physicians to prioritize further clinical examinations and provide personalized advice to patients based on risk contributors at the feature level.

By combining high accuracy, low computational cost, and transparent decision-making, **CluRF** stands out as a practical and reliable tool for real-world heart attack risk assessment, supporting both clinical workflows and patient engagement.

4.1 Limitation

Despite the promising performance and high interpretability of the proposed **CluRF** framework, several limitations should be acknowledged regarding its preprocessing strategy, data scale, and modeling assumptions.

First, our data preprocessing pipeline involved filtering out features with weak correlations—specifically, those with a Pearson correlation coefficient below 0.25 with all other variables. This step was intended to reduce noise, simplify model structure, and enhance predictive performance. However, this purely statistical criterion can overlook variables that have important clinical significance despite exhibiting low global correlations. Certain medical features, such as rare symptoms or interaction effects, may not be strongly correlated with others, but could still play a key role in specific subgroups of patients. The exclusion of such features could compromise the clinical completeness of our model and reduce its acceptance by domain experts.

Second, all datasets used in our study—as shown in Table 2—are relatively small in size. After clustering, the data within each subgroup becomes even more limited. For example, applying k-means with 5 or 6 clusters reduces the average number of training samples per cluster to below 200 in most cases. While clustering improves personalization and captures latent heterogeneity, this also increases the risk of overfitting, particularly when training separate Random Forest models within each cluster. The smaller the subgroup, the more sensitive the model becomes to noise and sampling variance, especially in imbalanced or sparse settings.

Third, our current design assumes that the number of clusters k is manually pre-specified and fixed across datasets. This rigid configuration may not reflect the true underlying structure of different populations and can lead to either under-segmentation or over-fragmentation. A more dynamic approach to determining cluster granularity could improve both model performance and robustness.

Furthermore, while **CluRF** demonstrates strong performance in controlled experimental settings, several deployment challenges remain unaddressed. These include integration complexities with diverse Electronic Health Record (EHR) systems, potential model degradation due to data drift over time, and computational optimization requirements for real-time clinical decision support.

In summary, while **CluRF** offers strong empirical results, its reliance on correlation-based feature selection, small sample sizes within clusters, and static clustering configurations highlight key areas where future improvements are needed.

4.2 Future Work

Looking ahead, we identify two promising directions to further enhance the applicability of **CluRF** in real-world clinical environments.

First, although our model integrates SHAP explanations to provide both global and individualized interpretability, the current outputs are presented as

static plots and numerical values. In future work, we plan to develop an interactive user interface that combines model predictions, feature attribution visualizations, and patient-specific summaries. Such an interface would support clinical decision-making by enabling physicians to quickly interpret the key drivers behind risk scores and explain them clearly to patients.

Second, all experiments in this study were conducted on public datasets, which, while widely used, are limited in size and diversity. To validate the robustness and generalizability of **CluRF**, future research should involve multi-center clinical datasets and real-world patient records. This would allow a more comprehensive evaluation of the model's performance across institutions, demographics, and healthcare settings. Moreover, integrating **CluRF** into electronic health record (EHR) systems would be an important step toward clinical deployment and adoption.

Future research will address key deployment challenges for real-world clinical implementation. This includes developing standardized APIs for seamless EHR integration across major healthcare platforms (Epic, Cerner, Allscripts), implementing continuous monitoring systems to detect and handle data drift as patient populations evolve, and optimizing computational efficiency to ensure real-time feasibility in resource-constrained clinical environments. Additionally, we plan to conduct extensive clinical validation studies to evaluate CluRF's performance in diverse healthcare settings and establish regulatory compliance pathways for clinical decision support system deployment.

References

1. Edelman, E.R., Rogers, C.: Pathobiologic responses to stenting. *Am. J. Cardiol.* **81**(7), 4E-6E (1998)
2. Wardlaw, J.M., Murray, V., Berge, E., Del Zoppo, G.J.: Thrombolysis for acute ischaemic stroke. *Cochrane Database of Systematic Reviews*, no. 7 (2014)
3. Wilson, P.W., D'Agostino, R.B., Levy, D., Belanger, A.M., Silbershatz, H., Kannel, W.B.: Prediction of coronary heart disease using risk factor categories. *Circulation* **97**(18), 1837–1847 (1998)
4. Karmali, K.N., Goff, D.C., Ning, H., Lloyd-Jones, D.M.: A systematic examination of the 2013 ACC/AHA pooled cohort risk assessment tool for atherosclerotic cardiovascular disease. *J. Am. Coll. Cardiol.* **64**(10), 959–968 (2014)
5. Cortes, C., Vapnik, V.: Support-vector networks. *Mach. Learn.* **20**, 273–297 (1995)
6. Breiman, L.: Random forests. *Mach. Learn.* **45**, 5–32 (2001)
7. Peterson, L.E.: K-nearest neighbor. *Scholarpedia* **4**(2), 1883 (2009)
8. Rosenblatt, F.: The perceptron: a probabilistic model for information storage and organization in the brain. *Psychol. Rev.* **65**(6), 386 (1958)
9. Wang, J., Xue, Q., Zhang, C.W., Wong, K.K.L., Liu, Z.: Explainable coronary artery disease prediction model based on AutoGluon from AutoML framework. *Front. Cardiovasc. Med.* **11**(1), 1360548 (2024)
10. Skouteli, C., Prenzias, N., Kakas, A., Pattichis, C.S.: Explainable AI modeling in the prediction of cardiovascular disease risk. *Stud. Health Technol. Inf.* **301**(1), 123–127 (2024)
11. Conroy, R.M., et al.: Estimation of ten-year risk of fatal cardiovascular disease in Europe: the SCORE project. *Eur. Heart J.* **24**(11), 987–1003 (2003)

12. Hippisley-Cox, J., et al.: Predicting cardiovascular risk in England and Wales: prospective derivation and validation of QRISK2. *BMJ* **336**(7659), 1475–1482 (2008)
13. D’Agostino, R.B., Grundy, S., Sullivan, L.M., Wilson, P., C.R.P. Group, et al.: Validation of the Framingham coronary heart disease prediction scores: results of a multiple ethnic groups investigation. *Jama* **286**(2), 180–187 (2001)
14. Sullivan, L.M., Massaro, J.M., D’Agostino, R.B., Sr.: Presentation of multivariate data for clinical use: the Framingham Study risk score functions. *Stat. Med.* **23**(10), 1631–1660 (2004)
15. Canedo, C., Fernández-Santamónica, A., Larriba, Y., Fernández, I., Rueda, C.: Heart attack outcome predictions using FMM models. *Comput. Cardiol.* **50**, 1–4 (2023)
16. Garg, A., Garg, N.B., Bansal, M.: Revolutionizing heart attack prevention: machine learning models in smart healthcare. *SN Comput. Sci.* **6**(2), 1–14 (2025)
17. Salah, M., Aljanabi, S., Mohammed, H., Jasim, H.: Hybridization of neural networks and sine cosine algorithm for an optimal neural network architecture. *J. Comput. Sci. Intell. Technol.* (2024, in press)
18. Dritsas, E., Trigka, M.: Application of deep learning for heart attack prediction with explainable artificial intelligence. *Computers* **13**(10), 244 (2024)
19. Nawar, A.K., Jebur, S.A., Jawad, H.A., Jahefer, M.M., Alkhazraji, L., Hussain, A.J.: Heart attack prediction by integrating independent component analysis with machine learning classifiers. *Int. J. Adv. Comput. Sci. Appl.* (2024, accepted/In press)
20. Luo, R., et al.: BioGPT: generative pre-trained transformer for biomedical text generation and mining. *Briefings Bioinf.* **23**(6), bbac409 (2022)
21. Singhal, K., et al.: Toward expert-level medical question answering with large language models. *Nat. Med.* **31**(3), 943–950 (2025)
22. Xie, J., Wang, Z., Yu, Z., Ding, Y., Guo, B.: Prototype learning for medical time series classification via human-machine collaboration. *Sensors* **24**(8), 2655 (2024)
23. Rezk, N.G., Alshathri, S., Sayed, A., El-Din Hemdan, E., El-Behery, H.: XAI-augmented voting ensemble models for heart disease prediction: a SHAP and LIME-based approach. *Bioengineering* **11**(10), 1016 (2024)
24. Pearl, J.: *Causality: Models, Reasoning, and Inference*, 2nd edn. Cambridge University Press, Cambridge (2009)
25. Peters, J., Janzing, D., Schölkopf, B.: *Elements of Causal Inference: Foundations and Learning Algorithms*. MIT Press, Cambridge (2017)
26. Ganie, S.M., Pramanik, P.K.D., Zhao, Z.: Ensemble learning with explainable AI for improved heart disease prediction based on multiple datasets. *Sci. Rep.* **15**(1), 13912 (2025)